National University of Computer and Emerging Sciences

Karachi Campus

Solution-Artificial Intelligence (AI-2002) (Spring 2026)

Question 01:

1

Admissible heuristics: a heuristic is admissible if the estimated cost is never more than actual cost from the current node to the goal node.

Consistent heuristics: A heuristic is consistent if the cost from the current node to a successor node plus the estimated cost from the successor node to the goal is less than or equal to estimated cost from the current node to the goal

2.

Feature	Depth-Limited Search (DLS)	Iterative Deepening (IDDFS)
Time Complexity	$O(b^l)$	$O(b^d)$
Space Complexity	$O(bl)$	$O(bd)$

3.

PEAS: Performance: patient safety & timely alerts; Environment: ICU patients and medical equipment; Actuators: alerts/ventilator controls; Sensors: heart rate, BP, oxygen monitors.
Environment type: Partially observable, stochastic, sequential, dynamic, continuous, multi-agent.
Agent type: Learning utility-based agent (optimizes patient outcomes and improves from feedback)

4.

Hill Climbing suffers from local maxima (gets stuck at a suboptimal peak), plateaus (flat regions with equal heuristic values), and ridges (requires combined/diagonal moves not allowed by simple operators).
Local maxima can be handled using Random Restart or Simulated Annealing, which allow escaping suboptimal points. Plateaus and ridges can be mitigated by sideways moves, stochastic hill climbing, larger neighborhood operators, or probabilistic approaches like Simulated Annealing

Question 02: Part (a)

Solution

Strategy	Write Path from S→G	Total Path Cost (S-G)	Number of Nodes Expanded to reach Goal	Memory Consumption (E.g., High, Medium, Low)
UCS	A → C → F → Safe	10	6	Medium
Greedy Best-First Search	A → C → F → Safe	10	4	Medium
IDS	A → B → D → Safe	20	6–7	Low

Question 02: Part (b)

Solution

Iteration 0:

Misplaced tile: 5
h(n) = 1, g(n) = 0, f(n) = 1

Iteration 1: Successor Evaluation

Move	g(n)	h(n)	f(n)
Up	1	2	3
Down	1	2	3
Left	1	3	4
Right	1	2	3

Selected (tie-break): Right move

Iteration 2: Expansion of Selected Node

Move	g(n)	h(n)	f(n)
Up	2	3	5
Down	2	2	4

Final Solution Path:

Initial -> Down -> Right -> Goal
Total Path Cost = 2

Question 03: Part (a)

i) 4, because at most the drone can move to any of the four adjacent squares.

ii) N^2C, because the drone's position can be one of N^2 squares. At any point in time, any combination of the p power cells and k destination squares will be visited, so we need one Boolean per power cell and one Boolean per destination square. The drone's current battery charge will be one of C different values. Thus, it gives a size of: $C\ N^2\ 2^{p+k}$

iii) BFS tree search, BFS graph search, UCS tree search, UCS graph search

Question 03: Part (b)

Step1: Fitness Calculation

S1=	1111010101	=7
S2=	0111000101	=5
S3=	1110110101	=7
S4=	0100010011	=4
S5=	1110111101	=8
S6=	0100110000	=3
	Total	34 → Fitness of All

Step 2: Crossover
S1: 11**10110101**
S3: 11**11010101**
S4: 01000**10000**
S6: 01001**10011**

Step 3: Mutation
S1: 111011**1**101
S3: 1111**1**101**1**1
S4: 0100000010
S6: 0100110**1**11

Step 3: Calculate Fitness again
S1: 1110111101=8
S2: 0111000101 =5
S3: 1111110111=9
S4: 0100000010=2
S5: 1110111101=8
S6: 0100110**1**11=6

Total= 37

Since Fitness is increased then students need to perform one more step and then stop